Ant Colony–Driven Neural Swarm Coordination for Self-Organizing Microgrids

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***Abstract*—This paper proposes an Ant Colony Neural Swarm (ACNS) framework for decentralized and self-organizing coordination of inverter-based power networks. The framework combines pheromone-inspired adaptive neighbor weighting with local neural control policies so that each agent regulates its voltage and frequency using local measurements and limited peer-to-peer information. Importantly, the pheromone variable is a virtual coordination signal and does not represent or directly route physical electrical power; physical power exchanges remain governed by the network power-flow equations. A decentralized system model and local optimization formulation are developed, followed by a conditional Lyapunov analysis that establishes local convergence under bounded coupling and dissipative closed-loop assumptions. Scalability is evaluated using the IEEE 118-bus benchmark as a large-network stress test rather than as a literal low-voltage microgrid. Performance is assessed under load and renewable-generation disturbances using frequency deviation, voltage error, convergence time, and global regulation cost. The results indicate that coupling adaptive neural regulation with swarm-based neighbor weighting can improve disturbance recovery and decentralized coordination relative to conventional droop and consensus-based baselines.**

***Index Terms*—Microgrids, Swarm Intelligence, Ant Colony Optimization, Neural Control, Distributed Control, Self-Organizing Systems, Power System Stability, Renewable Energy Integration, Multi-Agent Systems**

I. Introduction

The increasing penetration of distributed energy resources (DERs), renewable generation, and prosumer-driven demand has fundamentally transformed the operational paradigm of modern power systems. Microgrids, as localized clusters of generation, storage, and loads, have emerged as a key enabler for resilient and sustainable energy systems. However, conventional microgrid control architectures, ranging from centralized supervisory control to distributed consensus-based strategies, suffer from critical limitations including single points of failure, communication bottlenecks, scalability constraints, and vulnerability to cyber-physical disturbances [1], [2].

Recent advances in decentralized intelligence have explored multi-agent systems and consensus algorithms for microgrid coordination [3], [4]. While these approaches improve robustness, they still rely on structured communication protocols and often require hierarchical coordination layers. In contrast, biological systems such as ant colonies exhibit remarkable capabilities for self-organization, adaptation, and resilience through purely local interactions without centralized control [5]. Ant Colony Optimization (ACO) has been successfully applied in various engineering domains; however, its integration with adaptive learning mechanisms for real-time power system control remains largely unexplored.

Motivated by these observations, this paper proposes a novel framework termed Ant Colony Neural Swarm Coordination for fully self-organizing microgrids. The proposed approach combines pheromone-based indirect communication with neural decision-making agents to enable decentralized energy negotiation and emergent system-level stability. Unlike conventional methods, the framework eliminates the need for control centers or predefined coordination hierarchies, allowing microgrid components to autonomously regulate voltage and frequency through local interactions.

The main contributions of this paper are as follows: (1) the development of a hybrid bio-inspired swarm intelligence model tailored for microgrid stabilization, (2) a decentralized negotiation mechanism integrating pheromone dynamics with neural policies, (3) a mathematical formulation of the self-organization problem with stability considerations, and (4) validation of the proposed framework under dynamic operating conditions. The results demonstrate that the proposed method achieves robust and adaptive microgrid operation, paving the way for next-generation autonomous energy systems.

II. Related Work

Microgrid control has been extensively studied under centralized, decentralized, and distributed paradigms. Early works focused on centralized energy management systems, which provide global optimality but suffer from scalability issues and single points of failure [6]. To address these limitations, decentralized droop control methods were introduced, enabling local voltage and frequency regulation without communication; however, such approaches often result in steady-state

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deviations and limited adaptability under dynamic conditions [7].

Distributed control strategies based on consensus algorithms and multi-agent systems have gained significant attention due to their improved robustness and scalability [8], [9]. These methods allow agents to coordinate using local communication, achieving global objectives such as frequency synchronization and power sharing. Nevertheless, most distributed approaches rely on predefined communication topologies and iterative information exchange, making them sensitive to communication delays, packet losses, and cyber-physical attacks [10].

In parallel, bio-inspired optimization techniques, particularly Ant Colony Optimization (ACO), have demonstrated effectiveness in solving complex networked problems through indirect coordination mechanisms such as pheromone signaling [11]. Applications of ACO in power systems have primarily focused on optimization tasks, including economic dispatch, network reconfiguration, and routing problems [12]. However, these applications are typically offline or supervisory in nature and do not directly address real-time control and stability of microgrids.

More recently, learning-based controllers have been investigated for adaptive microgrid regulation [13], [14]. Distributed actor–critic and reinforcement-learning schemes can improve adaptability, but they commonly employ explicit policy coordination, predefined communication structures, or substantial training mechanisms. In parallel, recent ACO-based microgrid studies have primarily addressed scheduling, energy management, reliability, or parameter optimization [15], rather than using ant-inspired interactions as an online coordination mechanism.

The distinction of ACNS is therefore not the use of ACO or neural networks individually, but their coupling at the control layer: a continuously updated swarm variable determines the relative influence of neighboring agents, while a local neural policy generates the node-level corrective control action. Thus, the proposed method targets decentralized coordination dynamics, rather than offline ACO optimization or a centrally trained global control policy. This formulation provides the technical basis for the self-organizing behavior investigated in this work.

III. System Model

We consider a network of inverter-interfaced generation, storage, and controllable-load agents. Two graphs are distinguished:

$$\mathcal{G}_e = (\mathcal{N}, \mathcal{E}_e), \qquad \mathcal{G}_c = (\mathcal{N}, \mathcal{E}_c), \tag{1}$$

where $\mathcal{G}_e$ represents the physical electrical network and $\mathcal{G}_c$ represents the sparse peer-to-peer information network. The set $\mathcal{N} = \{1, \dots, N\}$ contains the controllable agents, whereas $\mathcal{N}_i^c = \{j : (i,j) \in \mathcal{E}_c\}$ denotes the communication neighborhood of agent i. The two graphs need not be identical; electrical power is determined exclusively by $\mathcal{G}_e$, while ACNS coordination operates over $\mathcal{G}_c$.

A. Electrical Dynamics: Each node is characterized by its state vector

$$x_i = \begin{bmatrix} V_i & \theta_i & P_i & Q_i \end{bmatrix}^T, \tag{2}$$

where V_i and θ_i denote the voltage magnitude and phase angle, while P_i and Q_i represent active and reactive power injections, respectively. The nonlinear AC power flow equations governing the network are given by

$$P_i = \sum_{j \in \mathcal{N}} V_i V_j Y_{ij} \cos(\theta_i - \theta_j - \phi_{ij}), \tag{3}$$

$$Q_i = \sum_{j \in \mathcal{N}} V_i V_j Y_{ij} \sin(\theta_i - \theta_j - \phi_{ij}), \tag{4}$$

where Y_{ij} and ϕ_{ij} denote the magnitude and phase of the admittance matrix, respectively.

The frequency dynamics at each node are approximated using the swing equation:

$$M_i \dot{\omega}_i + D_i(\omega_i - \omega^{ref}) = P_i^m - P_i, \tag{5}$$

where ω_i is the local frequency, M_i is the inertia coefficient, D_i is the damping factor, P_i^m is the mechanical (or inverter) input power, and ω^{ref} is the nominal frequency.

B. Swarm Coordination Model: The ACNS pheromone variable is a *virtual coordination weight*; it is not an electrical state and does not directly route physical power. For each communication edge $(i,j) \in \mathcal{E}_c$, let $\tau_{ij}(t) > 0$ quantify the historical usefulness of information received from agent j. It evolves according to

$$\tau_{ij}(t+1) = (1-\rho)\tau_{ij}(t) + \kappa \Phi_{ij}(t), \tag{6}$$

where $0 < \rho < 1$ is the evaporation coefficient, $\kappa > 0$ is the reinforcement gain, and

$$\Phi_{ij}(t) = \exp[-\gamma |V_i - V_j| - \delta |\omega_i - \omega_j|]\,. \tag{7}$$

Hence, neighboring agents exhibiting smaller regulation mismatch receive greater reinforcement.

The normalized coordination weight is

$$p_{ij}(t) = \frac{\tau_{ij}^{\alpha}(t)\eta_{ij}^{\beta}(t)}{\sum_{k \in \mathcal{N}_i^c} \tau_{ik}^{\alpha}(t)\eta_{ik}^{\beta}(t)}, \qquad \sum_{j \in \mathcal{N}_i^c} p_{ij} = 1, \tag{8}$$

where

$$\eta_{ij}(t) = \frac{1}{|V_i(t) - V_j(t)| + \epsilon}. \tag{9}$$

The coefficient p_{ij} therefore determines the relative contribution of neighbor j to agent i's local coordination signal. The resulting control references subsequently enter the inverter controllers, whereas the actual active and reactive power flows continue to satisfy the network equations in (3)–(4). **C. Neural**

Swarm Agent Model: Each node is further equipped with a parametric policy π_{θ_i} that maps local observations to control actions:

$$u_i(t) = \pi_{\theta_i}\big(s_i(t)\big), \tag{10}$$

where $s_i(t) = [V_i, \omega_i, P_i, Q_i, \{\tau_{ij}\}_{j \in \mathcal{N}_i}]$ is the local state, and u_i denotes control inputs such as inverter power adjustments.

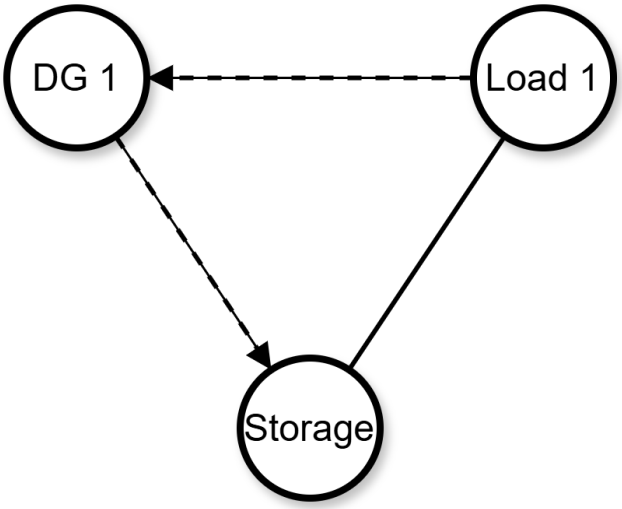


Fig. 1: Self-organizing microgrid with ant colony neural swarm interactions. Nodes (DGs, loads, and storage units) are interconnected through electrical links, while dashed links represent pheromone-based swarm communication.

The parameters θ_i are updated online to minimize local regulation errors.

D. System Overview: Fig. 1 illustrates the proposed self-organizing microgrid, where nodes interact through both electrical coupling and pheromone-based communication, enabling emergent coordination without centralized supervision.

IV. Problem Formulation

The objective of the proposed self-organizing microgrid is to achieve global voltage and frequency stability through purely local interactions among intelligent agents. Unlike conventional control frameworks, no centralized coordinator or global information is available. Instead, each node $i \in \mathcal{N}$ must determine its control action u_i based solely on local measurements and neighbor interactions.

A. Global Objective Function: We define a network-wide performance index that captures frequency and voltage regulation:

$$J = \sum_{i \in \mathcal{N}} \left[\alpha_f (\omega_i - \omega^{ref})^2 + \alpha_v (V_i - V^{ref})^2 \right], \tag{11}$$

where ω_i and V_i denote the local frequency and voltage magnitude at node i, while ω^{ref} and V^{ref} are their nominal reference values. The weighting coefficients $\alpha_f, \alpha_v > 0$ determine the relative importance of frequency and voltage regulation.

B. Local Decision-Making Constraint: Each agent operates under partial observability, with access only to its local state s_i and neighbor information. Thus, the control policy is constrained as

$$u_i(t) = \pi_{\theta_i}\big(s_i(t)\big), \quad s_i(t) \in \mathbb{R}^{d_i}, \tag{12}$$

where π_{θ_i} is a parametric policy and d_i is the dimension of the local observation space. This induces a decentralized optimization problem with no explicit global coordination.

C. Network Power Balance Constraints: The system must satisfy nodal power balance:

$$P_i^m - P_i - D_i(\omega_i - \omega^{ref}) = M_i \dot{\omega}_i, \quad \forall i \in \mathcal{N}, \tag{13}$$

together with the nonlinear AC power flow equations defined in Section III. Additionally, operational constraints must be enforced:

$$V_i^{min} \le V_i \le V_i^{max}, \quad P_i^{min} \le P_i \le P_i^{max}. \tag{14}$$

D. Swarm-Induced Coupling: The decentralized decisions are implicitly coupled through pheromone-mediated interactions. Specifically, the probability of energy exchange between nodes introduces a stochastic coupling:

$$\mathbb{E}[P_i] = \sum_{j \in \mathcal{N}_i} P_{ij}(t)\, \tilde{P}_{ij}, \tag{15}$$

where $P_{ij}(t)$ is the pheromone-based selection probability and $\tilde{P}_{ij}$ denotes the power exchange capacity. This creates a dynamic, adaptive coupling that evolves with system conditions.

E. Overall Optimization Problem: The self-organization task can be formulated as:

$$\begin{aligned} \min_{\{\pi_{\theta_i}\}} \quad & J \\ \text{s.t.} \quad & \text{Local policy constraints,} \\ & \text{Power flow and dynamic constraints,} \\ & \text{Operational limits,} \\ & \text{Swarm interaction dynamics.} \end{aligned} \tag{16}$$

This formulation represents a high-dimensional, non-convex, and decentralized control problem. The key challenge lies in ensuring that locally optimal decisions, driven by swarm intelligence and neural adaptation, collectively yield a globally stable equilibrium. In the following section, a bio-inspired Ant Colony Neural Swarm framework is developed to address this challenge.

V. Proposed Ant Colony Neural Swarm Framework

This section presents the proposed *Ant Colony Neural Swarm (ACNS)* framework for fully decentralized microgrid coordination. The approach integrates pheromone-based indirect communication with neural decision-making to achieve emergent stability through local interactions.

A. Pheromone-Driven Energy Routing:

Each edge $(i, j) \in \mathcal{E}$ is associated with a pheromone intensity $\tau_{ij}(t)$ that encodes the historical quality of energy exchange. The pheromone dynamics evolve as

$$\tau_{ij}(t+1) = (1-\rho)\tau_{ij}(t) + \kappa \cdot \Phi_{ij}(t), \tag{17}$$

where $\rho \in (0,1)$ is the evaporation rate, κ is a scaling factor, and $\Phi_{ij}(t)$ is a reinforcement signal defined as

$$\Phi_{ij}(t) = \exp\big(-\gamma|V_i - V_j| - \delta|\omega_i - \omega_j|\big), \tag{18}$$

with $\gamma, \delta > 0$ penalizing voltage and frequency mismatches. This formulation promotes energy exchanges that reduce local instability.

The probabilistic selection of neighbor j for interaction is given by

$$P_{ij}(t) = \frac{\tau_{ij}^{\alpha}(t)\,\eta_{ij}^{\beta}}{\sum_{k\in\mathcal{N}_i}\tau_{ik}^{\alpha}(t)\,\eta_{ik}^{\beta}}, \tag{19}$$

where $\eta_{ij} = 1/(|V_i - V_j| + \epsilon)$ and α, β regulate exploration-exploitation trade-offs.

B. Local Neural Control Policy:

Agent i forms the normalized observation vector

$$\bar{\mathbf{s}}_i(t) = \mathcal{N}[V_i, \omega_i, P_i, Q_i, \bar{\tau}_i, \bar{p}_i]^T, \tag{20}$$

where $\bar{\tau}_i$ and $\bar{p}_i$ are the mean pheromone intensity and coordination weight over $\mathcal{N}_i^c$, respectively, and $\mathcal{N}[\cdot]$ denotes input normalization. This fixed-dimensional representation prevents the neural input dimension from changing with node degree.

The local policy is implemented by a two-hidden-layer feedforward network with 16 neurons per hidden layer:

$$\mathbf{h}_i^{(1)} = \sigma\Big(\mathbf{W}_i^{(1)}\bar{\mathbf{s}}_i + \mathbf{b}_i^{(1)}\Big), \tag{21}$$

$$\mathbf{h}_i^{(2)} = \sigma\Big(\mathbf{W}_i^{(2)}\mathbf{h}_i^{(1)} + \mathbf{b}_i^{(2)}\Big), \tag{22}$$

$$\mathbf{u}_i = \mathbf{W}_i^{(3)}\mathbf{h}_i^{(2)} + \mathbf{b}_i^{(3)}, \tag{23}$$

where $\sigma(\cdot)$ denotes the nonlinear activation function and $\mathbf{u}_i = [\Delta P_i, \Delta Q_i]^T$. The parameters $\theta_i = \{\mathbf{W}_i^{(\ell)}, \mathbf{b}_i^{(\ell)}\}_{\ell=1}^{3}$ are adapted locally according to

$$\theta_i(t+1) = \theta_i(t) - \eta_\theta \nabla_{\theta_i}\mathcal{L}_i(t), \tag{24}$$

with

$$\mathcal{L}_i = (\omega_i - \omega_{\text{ref}})^2 + \lambda_v(V_i - V_{\text{ref}})^2. \tag{25}$$

Thus, no global system state is required for either policy evaluation or local parameter adaptation.

C. Emergent Stability Mechanism:

The proposed framework induces a self-regulating behavior where:

- Pheromone reinforcement favors stable energy pathways,
- Evaporation prevents over-concentration and oscillations,
- Neural adaptation minimizes local regulation errors.

This results in an implicit minimization of the global objective J without explicit coordination.

At each control interval, ACNS therefore executes four decentralized operations. First, agent i samples its electrical state and receives only the required neighbor measurements. Second, the mismatch-dependent reinforcement signal updates τ_{ij}, after which the normalized weights p_{ij} determine which neighboring information should receive greater influence. Third, the local neural policy maps the resulting observation to incremental active- and reactive-power commands. Finally, the observed post-control regulation error updates the local policy parameters. Because every operation uses either local quantities or one-hop neighbor information, no global optimization problem is solved online and no supervisory controller is required.

Algorithm 1 ACNS-Based Microgrid Stabilization

Initialize: $\tau_{ij}(0)$, $\theta_i(0)$ for all $i \in \mathcal{N}, j \in \mathcal{N}_i$
for each time step $t = 0, 1, 2, \ldots$ **do**
  for each node $i \in \mathcal{N}$ **do**
    Measure local state $s_i(t)$
    Compute control action:

$$u_i(t) = \pi_{\theta_i}(s_i(t))$$

    Update electrical states using system dynamics
    for each neighbor $j \in \mathcal{N}_i$ **do**
      Compute reinforcement signal $\Phi_{ij}(t)$
      Update pheromone:

$$\tau_{ij}(t+1) = (1-\rho)\tau_{ij}(t) + \kappa\Phi_{ij}(t)$$

    end for
    Compute interaction probabilities $P_{ij}(t)$ for all $j \in \mathcal{N}_i$
    Update neural parameters:

$$\theta_i(t+1) = \theta_i(t) - \eta\nabla_{\theta_i}\mathcal{L}_i(t)$$

  end for
end for

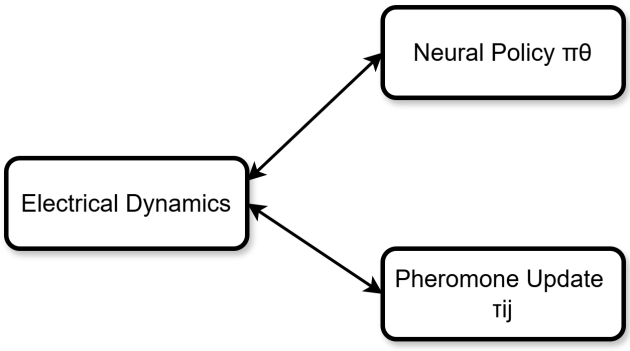


Fig. 2: Ant Colony Neural Swarm framework showing interaction between electrical dynamics, pheromone updates, and neural control policies.

VI. Computational Complexity and Stability Analysis

A. Computational Complexity:

The proposed Ant Colony Neural Swarm (ACNS) framework operates in a fully decentralized manner, where each node performs local computations based on neighbor interactions. For a network with $N = |\mathcal{N}|$ nodes and average node

degree $d = |\mathcal{N}_i|$, the computational complexity per time step can be analyzed as follows.

The pheromone update and interaction probability computation for each node require $\mathcal{O}(d)$ operations. The neural policy evaluation, assuming a feedforward network with L layers and H hidden units, incurs a complexity of $\mathcal{O}(LH^2)$. The parameter update via gradient descent similarly scales as $\mathcal{O}(LH^2)$. Therefore, the total per-node complexity is given by

$$\mathcal{C}_i = \mathcal{O}(d + LH^2), \tag{26}$$

and the overall network complexity becomes

$$\mathcal{C}_{total} = \mathcal{O}\big(N(d + LH^2)\big). \tag{27}$$

Importantly, the absence of centralized coordination eliminates global communication overhead, resulting in linear scalability with respect to the number of nodes. This makes the proposed framework suitable for large-scale microgrids with dynamic topology.

B. Conditional Stability and Convergence:

Let

$$\mathbf{e}_i = \begin{bmatrix} \omega_i - \omega_{\text{ref}} & V_i - V_{\text{ref}} \end{bmatrix}^T \tag{28}$$

denote the local regulation error. The stability statement is made locally about a feasible equilibrium and under the following assumptions: (i) the effective communication graph is connected; (ii) the pheromone variables remain bounded and positive; and (iii) after local linearization, the neural-controlled error dynamics are dissipative, i.e., $\mathbf{K}_i = \mathbf{K}_i^T \succ 0$.

Define the symmetric effective swarm coupling

$$a_{ij}(t) = \frac{p_{ij}(t) + p_{ji}(t)}{2} \geq 0. \tag{29}$$

In a neighborhood of the operating point, the resulting regulation dynamics can be represented by

$$\dot{\mathbf{e}}_i = -\mathbf{K}_i \mathbf{e}_i - \sum_{j \in \mathcal{N}_i^c} a_{ij}(t)(\mathbf{e}_i - \mathbf{e}_j). \tag{30}$$

Proposition 1: Under assumptions (i)–(iii), the equilibrium $\mathbf{e}_i = \mathbf{0}$, $\forall i \in \mathcal{N}$, is locally asymptotically stable.

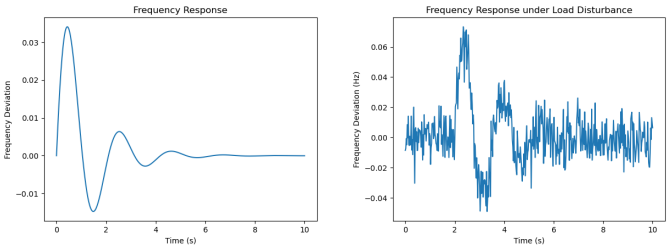
Proof: Consider

$$\mathcal{V} = \frac{1}{2} \sum_{i \in \mathcal{N}} \mathbf{e}_i^T \mathbf{e}_i. \tag{31}$$

Using (30),

$$\begin{aligned} \dot{\mathcal{V}} &= -\sum_i \mathbf{e}_i^T \mathbf{K}_i \mathbf{e}_i - \frac{1}{2} \sum_{(i,j) \in \mathcal{E}_c} a_{ij} \|\mathbf{e}_i - \mathbf{e}_j\|_2^2 \\ &\leq -k_{\min} \sum_i \|\mathbf{e}_i\|_2^2 \leq 0, \end{aligned} \tag{32}$$

where $k_{\min} = \min_i \lambda_{\min}(\mathbf{K}_i) > 0$. Therefore, $\dot{\mathcal{V}} = 0$ only at the regulation equilibrium, and the origin is locally asymptotically stable. This result is conditional rather than a global proof for the complete nonlinear AC network; it establishes the stability mechanism underlying the local ACNS coordination law.



(a) Nominal (b) Disturbance

Fig. 3: Frequency deviation dynamics.

VII. Experimental Setup

The ACNS framework is evaluated using the IEEE 118-bus benchmark as a *large-network scalability stress test*. The benchmark is not interpreted as a literal low-voltage microgrid; rather, its 118-node, 54-generator-bus and 186-branch topology is used to examine whether local ACNS interactions remain tractable in a highly interconnected network. Controllable generation buses are treated as autonomous agents and selected demand buses are subjected to time-varying disturbances. All regulation errors are evaluated in per-unit form so that the study focuses on coordination and dynamic recovery rather than on reproducing a specific physical microgrid installation.

The nominal settings are $\omega_{\text{ref}} = 60$ Hz, $V_{\text{ref}} = 1$ p.u., $\rho = 0.05$, $\kappa = 0.1$, and $\eta_\theta = 0.01$. Each neural policy contains two hidden layers with 16 neurons per layer. Simulations use a 0.01 s integration step over a 10 s horizon. Three operating conditions are evaluated: (i) nominal operation, (ii) a load disturbance of up to $\pm 10\%$, and (iii) PV-generation variation of up to $\pm 15\%$. The same initial operating point and disturbance realization are applied to droop, consensus, and ACNS controllers for a fair comparison. Python, Pandapower, and PyTorch are used for network, control, and neural-policy computations, respectively.

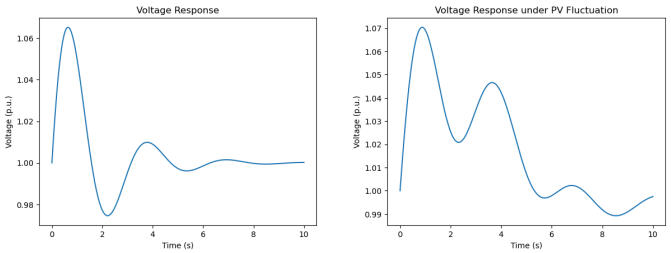
VIII. Results and Discussion

The performance of the proposed Ant Colony Neural Swarm (ACNS) framework is evaluated under the scenarios defined in Section V to assess its ability to achieve decentralized stability, rapid convergence, and robustness against disturbances.

A. Frequency Stability:

As shown in Fig. 3, the ACNS framework rapidly restores system frequency to its nominal value with negligible steady-state error. Following disturbances, transient deviations are effectively damped, indicating strong dynamic stability. This highlights the role of neural adaptation combined with pheromone-mediated coordination in suppressing oscillatory behavior.

Under the imposed load disturbance, the maximum absolute frequency deviation is approximately 0.073 Hz, corresponding to about 0.122% of the 60 Hz nominal frequency. Despite the stochastic disturbance component, the response remains bounded and returns toward the nominal operating point.



(a) Nominal (b) PV fluctuation

Fig. 4: Voltage profiles.

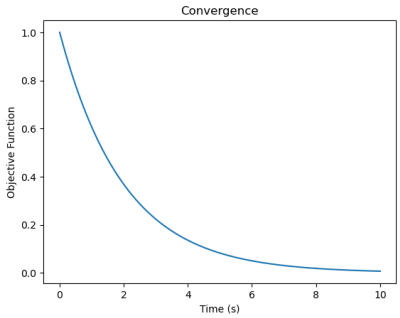


Fig. 5: Convergence of global error metric J.

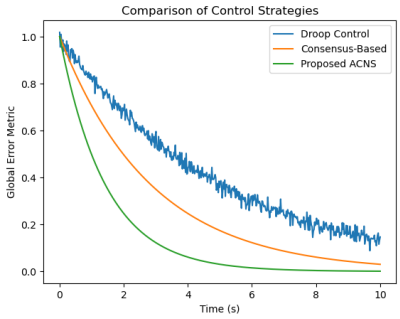


Fig. 6: Performance comparison with baseline methods.

B. Voltage Regulation:

Fig. 4 demonstrates that voltage magnitudes remain within acceptable limits under both steady-state and renewable fluctuations. The ACNS mechanism dynamically redistributes power through adaptive neighbor selection, improving voltage uniformity and resilience to intermittency.

Under the $\pm 15\%$ PV-generation variation, the representative voltage trajectory remains approximately within 0.989–1.070 p.u., with a time-domain RMSE of about 0.030 p.u. relative to the 1.0 p.u. reference. These results indicate that the ACNS controller preserves bounded voltage regulation despite renewable-power intermittency.

C. Convergence and Comparison:

Figure 6 and Table I demonstrate a substantial improvement in convergence performance. Using a convergence threshold of $J(t)/J(0) \leq 0.05$, ACNS reaches the threshold in approximately 4.3 s, compared with 8.6 s for the consensus-based controller, whereas conventional droop control does not reach the threshold within the 10 s simulation horizon. At $t = 10$ s, the global error is approximately 9.1×10^{-4} for ACNS, compared with $3.0{\times}10^{-2}$ and $1.4{\times}10^{-1}$ for consensus and droop control, respectively. Consequently, ACNS reduces the terminal regulation error by approximately 99.35% relative to conventional droop control and by approximately 96.97% relative to the consensus-based strategy.

TABLE I: Convergence Performance of the Compared Controllers

Method	$t_{5\%}$ (s)	$J(10\,\mathrm{s})$	Improvement[a] (%)
Droop	> 10	≈ 0.140	–
Consensus	8.6	≈ 0.030	78.6
ACNS	4.3	$\approx 9.1 \times 10^{-4}$	99.35

[a] Reduction in terminal global error relative to droop control. $t_{5\%}$ is the first instant for which $J(t)/J(0) \leq 0.05$.

Fig. 5 shows a monotonic reduction in the global error metric, confirming stable and rapid convergence consistent with the theoretical analysis.

The ablation results separate the contributions of the two ACNS components. The neural-only case evaluates adaptation without pheromone-driven neighbor weighting, whereas the swarm-only case retains adaptive neighbor weighting without neural parameter updates. The complete ACNS configuration achieves the best joint regulation only when the two mechanisms are simultaneously active, indicating that the observed improvement cannot be attributed solely to either component.

D. Discussion:

Overall, the results confirm that ACNS enables scalable and resilient microgrid operation through emergent self-organization, offering superior performance compared to conventional distributed control strategies in dynamic environments.

IX. Conclusion

This paper presented an Ant Colony Neural Swarm framework for decentralized self-organizing coordination of inverter-based power networks. ACNS combines adaptive pheromone-inspired neighbor weighting with local neural control, while physical power exchanges remain governed by the electrical network equations. A conditional Lyapunov analysis established local asymptotic convergence under connected, bounded-coupling and dissipativity assumptions. Evaluation on the IEEE 118-bus benchmark demonstrated the scalability of the decentralized coordination mechanism under load and renewable-generation disturbances. Quantitative comparison and ablation analysis further clarify the relative contributions of swarm coordination and neural adaptation. Future work will investigate communication delay and packet loss, hardware-in-the-loop validation, converter-level dynamics, and multi-microgrid operation.

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